

What Predicts a Movie's Financial Success?

DS2500 - Team Project Final Report

Team Members:

- [YOUR NAME] ([your_email]@northeastern.edu), solo project

Date: August 13, 2026

1. Introduction

Goal of the Project

This project looks at whether a movie's budget, runtime, and genre can predict whether it earns back its production budget, and how much revenue it is likely to make.

Problem Statement & Background

Making a movie is a big financial gamble. A single film can cost anywhere from \$50 million to well over \$200 million to produce, and a large share of movies never earn that money back. Studios usually have to make funding decisions without knowing for sure whether a movie will succeed, so a lot of that decision-making comes down to industry experience and gut feeling rather than hard evidence.

This project narrows that big industry problem down into one specific question we can actually test: using only information we would know before or at release, meaning budget, runtime, and genre, can we predict whether a movie will earn back its budget (a hit vs. a flop), and roughly how much revenue it will bring in? Answering this tells us how much of a movie's financial outcome comes from basic, easy-to-know factors, versus things this project doesn't measure, like marketing, cast, reviews, and timing.

Why This Matters

- Studios and investors take on real financial risk every time they greenlight a movie. Knowing which basic factors actually line up with that risk could help them weigh a project's financial risk against its creative or strategic value.

- Marketing and release scheduling decisions often assume a certain earning potential ahead of time. This project shows how much weight budget and genre actually deserve compared to other planning decisions.
- For anyone studying the business side of entertainment, this gives a real, measured answer to how much of a movie's financial fate comes down to simple, structural factors rather than harder to measure things like star power or plain luck.

2. Dataset

Dataset Overview

Source: TMDB 5000 Movie Dataset, published on Kaggle (kaggle.com/datasets/tmdb/tmdb-movie-metadata). It was originally put together from The Movie Database (TMDB), a movie information site kept up by its community.

Time period: the dataset covers movies going back to the early 1900s, up through around 2017, which is when this dataset was last updated. Most of the movies in it are from recent decades of studio and independent releases.

Size: 4,803 movies and 20 columns in the original file. After removing rows with missing financial or runtime data (explained in Section 3), 3,226 movies were left to work with.

Format: a single CSV file.

Data Collection Methodology

This data wasn't collected through a survey or any kind of sensor. TMDB pulled it together from studio-provided information and from a large group of volunteer editors, so it reflects whatever information those contributors happened to know or record for each movie. It isn't a carefully sampled record of every movie ever made.

Key Variables

Variable	Description	Type	Example
budget	Reported production budget in USD	Numeric	150000000
revenue	Total worldwide box office gross in USD	Numeric	623279547
runtime	Movie length in minutes	Numeric	148
genres	Raw list of genre tags (nested, needed parsing)	Text (nested)	[{"name": "Action"}, ...]
primary_genre	First genre listed, used as a simplified genre feature	Categorical	Action

release_date	Original theatrical release date	Date	2010-07-16
vote_average / vote_count	TMDB user rating and number of votes	Numeric	8.1 / 13752
hit_or_flop (derived)	1 if revenue $\geq$ budget, else 0	Binary	1
budget_log / revenue_log (derived)	log10 of budget / revenue, used to correct skew	Numeric	8.18

Ethical Considerations

- Privacy: this dataset is about movies and companies, not people, so there isn't much personal privacy risk here.
- Reporting bias: budget and revenue numbers are self-reported, and they're often missing for independent, foreign, or older movies. Removing those rows (Section 3) probably pushes the remaining data toward bigger, better documented movies.
- Inconsistent community data: genre order isn't always meaningful. We found real proof of this. Spider-Man 2 and Spider-Man 3 list different genres first (Action vs. Fantasy) even though they're very similar movies.
- Representation bias: this dataset only includes movies that were finished and released, so it can't tell us anything about projects that were pitched or started but never finished.

3. Methods

Data Preprocessing / Preparation

Several columns use zero to mean the information was never reported, not a real zero. A movie can't actually have a runtime of zero minutes. So we removed any row where budget, revenue, or runtime was zero, along with rows that had no votes or no release date, since keeping those would have made financial comparisons meaningless. This brought the dataset down from 4,803 to 3,226 movies.

Budget and revenue were both log transformed (base 10) to fix their strong right skew, since a small number of blockbuster movies are worth far more than a typical movie in this dataset (Figure 1). The genre column originally came in as a nested, string encoded list instead of plain text, so we parsed it down to a single genre per movie. We

used only the first genre listed as a simplification, though we found real evidence (Section 2) that this order isn't fully reliable, which we treat as a known limitation.

Our target label marks a movie as a hit (1) if its revenue was at least its budget, and a flop (0) otherwise. This target is imbalanced: about 76% of movies were hits and 24% were flops. That imbalance shaped a few of the methods below.

Exploratory Data Analysis

Early exploration confirmed that revenue and budget are both heavily right skewed (Figure 1). Hit rate also changes quite a bit by genre, from about 70% for Drama up to about 88% for Horror (Figure 2), and budget and revenue show a positive but noisy relationship (Figure 3): higher budget movies tend to earn more, but there's a lot of variation at every budget level.

Modeling & Evaluation

We used two classification methods for the hit or flop outcome, and one regression method to predict revenue directly. K-Nearest Neighbors predicts a movie's outcome by comparing it to the most similar movies already in the data. Logistic Regression works in a completely different way, fitting one equation across the whole dataset instead of comparing to nearby examples. Using two different methods let us check whether a result was just a quirk of one algorithm, or a real limitation of the features themselves.

We split the data into training (60%), validation (20%), and test (20%) sets. For KNN, we used the validation set to pick the number of neighbors (k) by testing odd values and choosing whichever one got the best F1 score on the flop class, not the best accuracy, since accuracy can be misleading when one outcome is much more common than the other. The test set was only used once, after every modeling decision was already made. Logistic Regression was tested on that same test set and features, with no extra tuning.

Separately, Linear Regression predicted log transformed revenue from the same three features, to see how much of a movie's financial outcome (not just hit or flop) these attributes explain. We used an 80/20 train test split and reported R squared and mean absolute error as our metrics.

4. Results & Conclusions

Finding 1: Revenue and budget are both heavily skewed.

A small number of blockbuster movies earn far more than a typical movie in this dataset. That's why both budget and revenue were log transformed before any further analysis (Figure 1). This is mostly a data preparation step, but it also means something real: financial success in this industry isn't spread out evenly. A small number of huge hits bring in a disproportionate share of all the revenue.

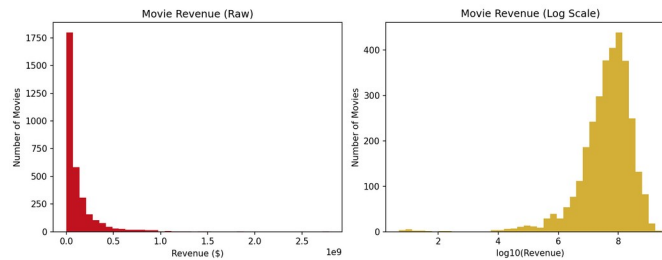


Figure 1: Movie revenue is extremely right-skewed on a raw dollar scale (left). A $\log_{10}$ transform (right) produces a far more usable, roughly bell-shaped distribution.

Finding 2: Genre and budget both show real, but partial, links to financial success.

Hit rate changes by about 18 percentage points across genres with a decent sample size, from about 70% for Drama up to about 88% for Horror (Figure 2). Budget and revenue are positively related, but that relationship is noisy, not tight (Figure 3). Plenty of high budget movies underperform, and some lower budget movies do much better than expected for their budget tier.

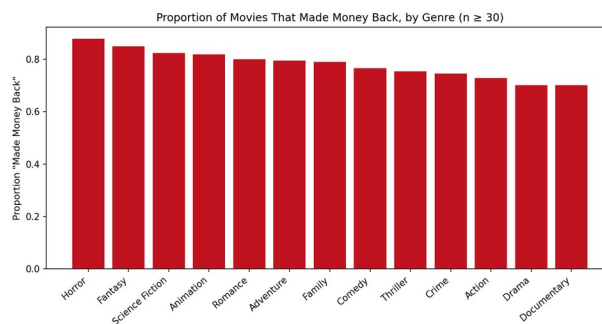


Figure 2: Hit rate by genre, limited to genres with at least 30 movies in the cleaned dataset. Horror and Fantasy movies were the most likely to earn back their budget. Drama and Documentary were the least likely.

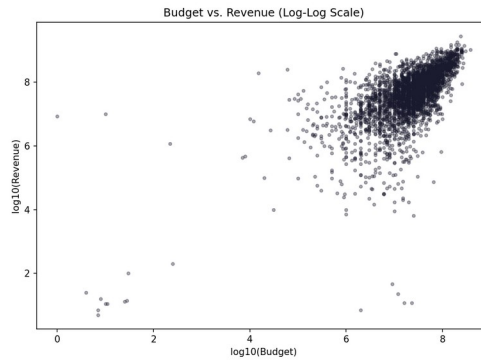


Figure 3: Budget vs. revenue on a log-log scale. The overall upward trend confirms a real relationship, but the wide spread at every budget level shows that budget alone doesn't tightly determine revenue.

Finding 3: A more accurate model was actually worse at the real task.

Logistic Regression had higher overall accuracy than KNN (75.7% vs. 71.8%), but its F1 score on the flop class, which is the metric we actually used to judge these models since flops are the harder and more useful outcome to predict, was exactly 0.00 (Figure 4). Logistic Regression never once correctly identified a flop in the test set. It just predicted the majority “hit” outcome for almost every movie, which happens to be right about three quarters of the time given how imbalanced this data is. KNN, even with lower accuracy, actually identified some real flops (F1 = 0.18). This is a clear example of why accuracy alone can be a misleading metric when your data is imbalanced, and why we judged our models by F1 on the minority class instead.

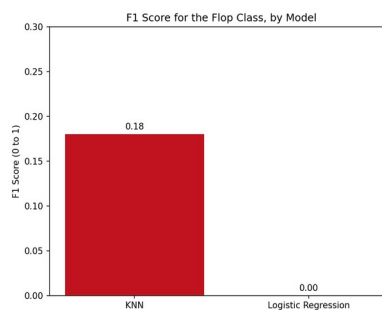


Figure 4: F1 score for the flop class, by model. Logistic Regression's higher overall accuracy comes with a complete failure to identify any flops.

Finding 4: Budget, runtime, and genre explain about 58% of the variation in revenue.

Our Linear Regression model, predicting log transformed revenue from budget, runtime, and genre, got an R squared of 0.577 on held out test data, with a mean absolute error of 0.42 in log10 units (roughly a 2.6 times typical error factor on the original revenue scale). Figure 5 shows this directly. Predicted and actual revenue trend upward together, but individual movies scatter pretty widely around the line of perfect prediction. This connects right back

to Findings 2 and 3. If these features only explain about half of what drives revenue, then no classifier built from those same features could realistically do much better at predicting hit or flop either.

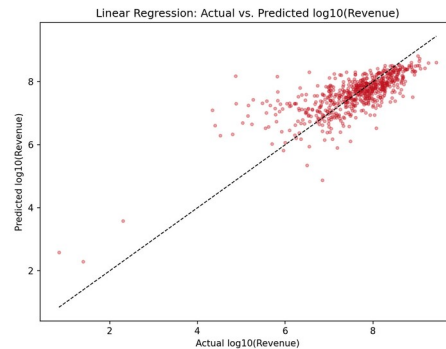


Figure 5: Actual vs. predicted $\log_{10}(\text{revenue})$ for the Linear Regression model. The dashed line represents a perfect prediction. Points trend upward together ($R^2 = 0.577$), but scatter widely around the line.

Model Performance Summary

Model	Metric	Value	Interpretation
KNN (k=5)	Accuracy	71.8%	Overall correct-guess rate on the test set
KNN (k=5)	Precision (flop)	0.31	Of movies predicted as flops, 31% actually were
KNN (k=5)	Recall (flop)	0.13	Only 13% of real flops were correctly caught
KNN (k=5)	F1 (flop)	0.18	Low overall ability to identify the minority class
Logistic Regression	Accuracy	75.7%	Higher accuracy than KNN, but see below
Logistic Regression	F1 (flop)	0.00	Never once correctly identified a flop
Linear Regression	R^2	0.577	Explains ~58% of the variation in log revenue
Linear Regression	MAE (log revenue)	0.42	Typical prediction error of roughly 2.6× the true value

Conclusions

Going back to our original question, can budget, runtime, and genre predict a movie's financial success, our analysis supports a partial answer: yes, but only somewhat. These features carry real, measurable signal (Figures 2, 3, and an R^2 of 0.577), but they're far from enough to reliably predict any single movie's hit or flop outcome. Our best classifier only caught 13% of true flops, and a model chosen purely for accuracy caught none at all. The clearest

takeaway is that basic pre-release information is a meaningful but incomplete signal of financial risk. Real risk assessment probably needs richer information than just budget, runtime, and genre.

5. Future Work & Limitations

Limitations of Current Analysis

- Data constraints: about a third of the dataset was removed because of missing financial figures, which probably skews the sample toward bigger, better documented movies.
- Methodological limitation: using only the first listed genre is a known simplification. The Spider-Man example confirmed it isn't fully reliable, and it loses information for movies that have more than one genre.
- Class imbalance: our target is imbalanced (76% hit, 24% flop). We handled this by judging with F1 on the minority class instead of accuracy, but we didn't apply class weighting or resampling, since those methods were outside what this course covered.
- Generalizability: our findings describe finished, released, mostly large movies, and shouldn't be assumed to apply to independent films or movies released after 2017.
- Assumptions: we define a hit strictly as revenue at or above budget, which ignores marketing, distribution costs, and other revenue that affects a movie's real profitability.

Future Research Directions

- Add features like cast and crew popularity, studio track record, and release timing, to see if they improve on the 58% of revenue variation we found here.
- Apply class imbalance correction methods to see if flop detection can improve without the big accuracy trade-off we saw in this project.
- Use all listed genres per movie instead of just the first, and add more recent movies to the dataset.

References

1. TMDB 5000 Movie Dataset. (2017). The Movie Database (TMDB), via Kaggle.

<https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>